Abstract

This optical heart rate sensor will result in the design and testing of an optical heart rate sensor using infrared light transmission through a finger, which is then measured on the other side with a phototransistor. From the sensor, the output will be amplified using an active high-pass and low-pass filter to correctly transmit the heart rate signal which lies between 40 BPM and 200 BPM. After this signal is filtered, another high-pass filter with a 2 Hz cutoff will be used to remove any additional DC-offset, and the output of this filter will be fed into a comparator with hysteresis. This will convert the filtered analog heartbeat signal into a digital signal that indicates whenever a heartbeat is detected through an LED. Overall, this project will demonstrate how infrared sensors, filtering, and analog to digital conversion can create the final optical heart rate sensor.

Components Used

- IR204 infrared LED, PT204-6B infrared phototransistor, LED, capacitors, resistors, LM324, LM339, AD2, and wires

Theory

The subsystems of this project can be split up into the following: an optical sensor system, a system of active filters, an analog to digital converter, and finally an indicator system (LED).

For the optical sensor system, the system operates by having an infrared LED shine through the user's finger, which will then be detected on the other side with a phototransistor. When blood travels between the transmitter and receiver, it slightly affects the amount of signal transmitted as well. The following equation describes the relation between the phototransistor voltage with the phototransistor current and emitter resistor:

$$v_{PT} = i_{PT} R_E$$

Additionally, the equation below describes the amount of infrared of LED current and its relationship with the LED resistor and diode forward voltage:

$$I_D = \frac{(V_{CC} - V_D)}{R_D}$$

The end result of the optical sensor subsystem is that there will be a small voltage spike whenever the user emits a heartbeat, allowing for an indication of the heartrate.

However, since the analog output of the optical subsystem is very small and contains noise and a DC offset, we must do additional filtering. The analog output of the system is placed through active high-pass and active low-pass filters. The high-pass filter removed DC offset while the low-pass filter removes high-frequency noise from the signal, leaving only the desired heart rate signal that lies between 40 BPM and 200 BPM. Additionally, each active filter stage will also amplify the signal so that it is easily readable. The cutoff frequencies for both filters can be calculated using the following equation:

$$f_c = \frac{1}{2\pi RC}$$

Additionally, the gains of each active filter stage will be multiplied with each other, giving a very high total system gain.

The last subsystem is the comparator with hysteresis, that converts the filtered analog signal into a digital output. The hysteresis voltage can be calculated by the following:

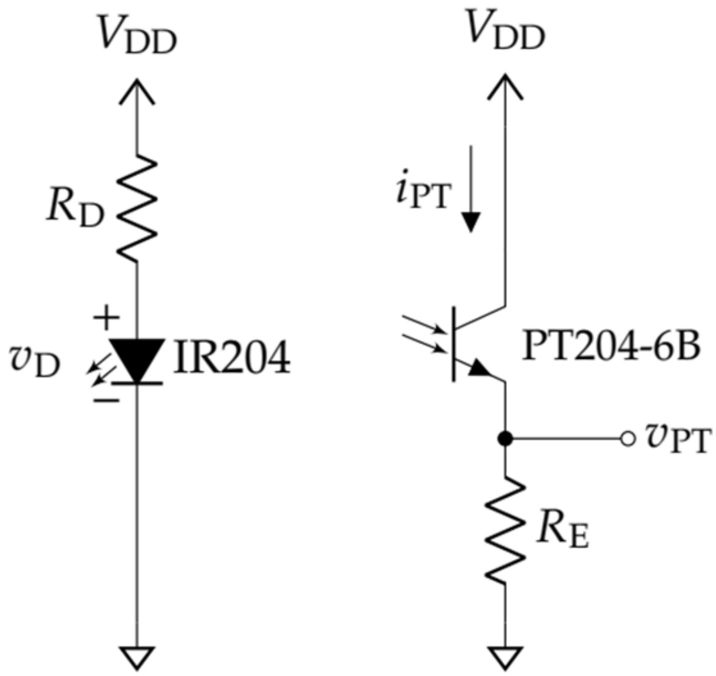
$$v_{hyst} = \frac{(V_{DD} - V_{SS})R_1}{R_1 + R_2}$$

The output of this hysteresis is what is tied to the final LED, which actually indicates the heartrate.

Design/Calculations

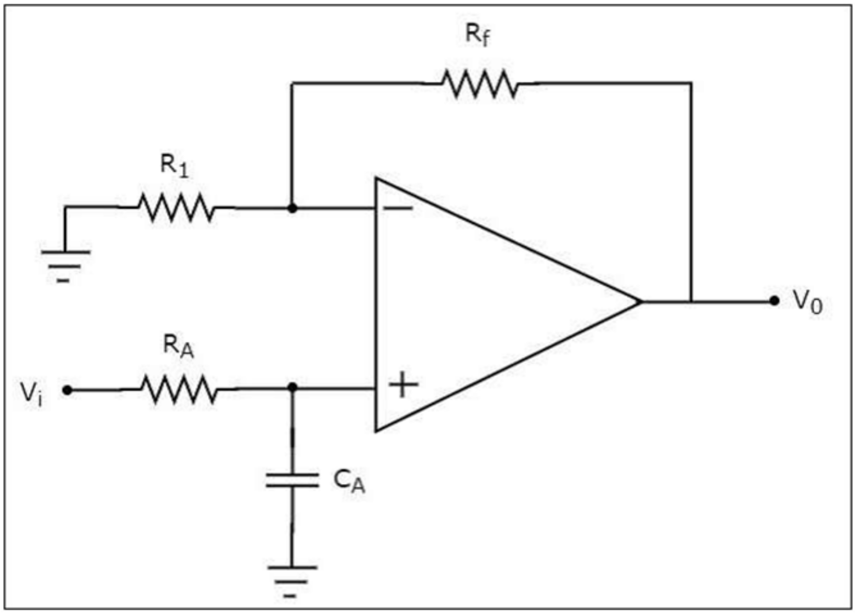
Using the above theory and equations, we come up with the following calculations for our design:

Optical Sensor:



For the transmitter, the typical voltage at 100mA is 1.4V, where the continuous forward current rated for our component is at 100mA. Therefore, we chose a 33 Ohm resistor for our value, which gives a 1.33V at the IR204 along with a 109mA current through the branch. Furthermore, setting up the receiver, we experimented with various resistor values until the voltage after the PT204-6B lies between 2-4V. By changing the resistor between the component and ground, we eventually get a resistance of ~3.4V with a resistor value of 56k. Refer to the final design topology for more context.

Active Filters:



For our active filters, we had to design and active low-pass and high-pass filter to allow only a range of heartbeats between 40BBPM and 200BPM to pass through.

For the high-pass filter, 40BPM translates to 0.67 Hz, but since we want to also include 40BPM within that range, we aim for 0.5 Hz. We go for a capacitor value of 10uF with a resistor value of 33k. Plugging this into the formula we get a frequency of:

$$f_c = \frac{1}{2\pi \cdot 33k \cdot 10uF} = 0.48\, Hz$$

Then for the low pass filter we need a BPM of 200, which translates into frequency of 3.33. Again, since we want to include values within that range, we aim for a frequency of 3.4 Hz. We use a capacitor value of 10uF and a resistor value of 4.7k. This gives a frequency of:

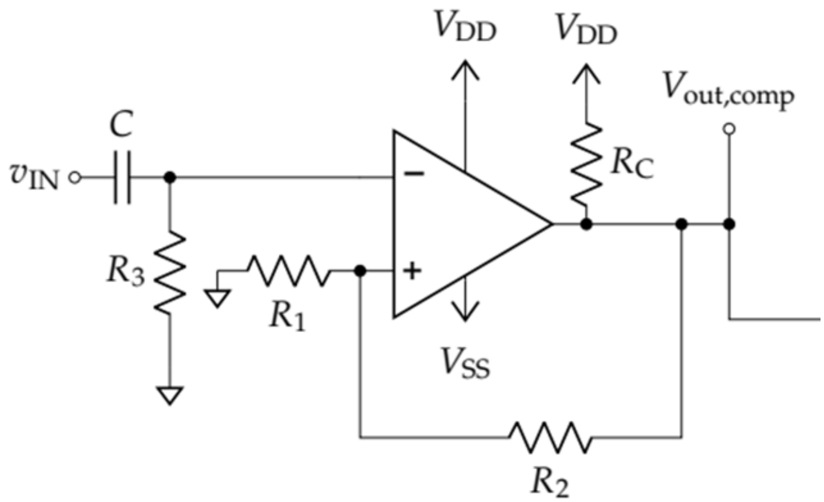
$$f_c = \frac{1}{2\pi \cdot 4.7k \cdot 10uF} = 3.38Hz$$

Then, for both filter stages, there is also an amplification stage that follows. For both op-amps, we use resistor values of R_1 = 10K and R_2 = 150K. Using the non-inverting gain formula:

$$A_v = 1 + \frac{R_1}{R_2}$$

We get a gain of 16 for both stages. These values multiply with each other (16 * 16 = 256) to give us a total gain of 256, which matches what the lab manual specifies for our design.

Analog to Digital Converter:



The first thing that needs to be calculated for this step is a high-pass filter with a 2-Hz cut off frequency. To design this, we use a 0.1 uF capacitor with an 820k resistor. This results in a frequency of:

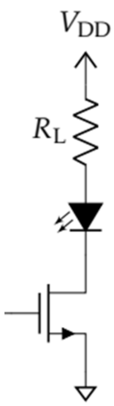
$$f_c = \frac{1}{2\pi \cdot 820k \cdot 0.1uF} = 1.94$$

Secondly, the comparator also requires a R_1 and R_2 value to keep v_hyst between the heartbeat noise and actual signal. To find these, we played around with two 1M potentiometers to get values of R_1 = 100k and R_2 = 720k. This gives us a v_hyst calculation of:

$$v_{hyst} = (V_{DD} - V_{ss})\left(\frac{R_1}{R_1 + R_2}\right) = 1.2$$

Lastly, we have a 1k pullup resistor to prevent noise in the final signal.

Indicator:



Lastly, for the indiciator, we use an LED with an NMOS to indicate the final digital signal. We choose a resistor of 150 to maximize current flowing into the LED:

Results:

Below is our final circuit diagram, with all of the previously described calculated values:

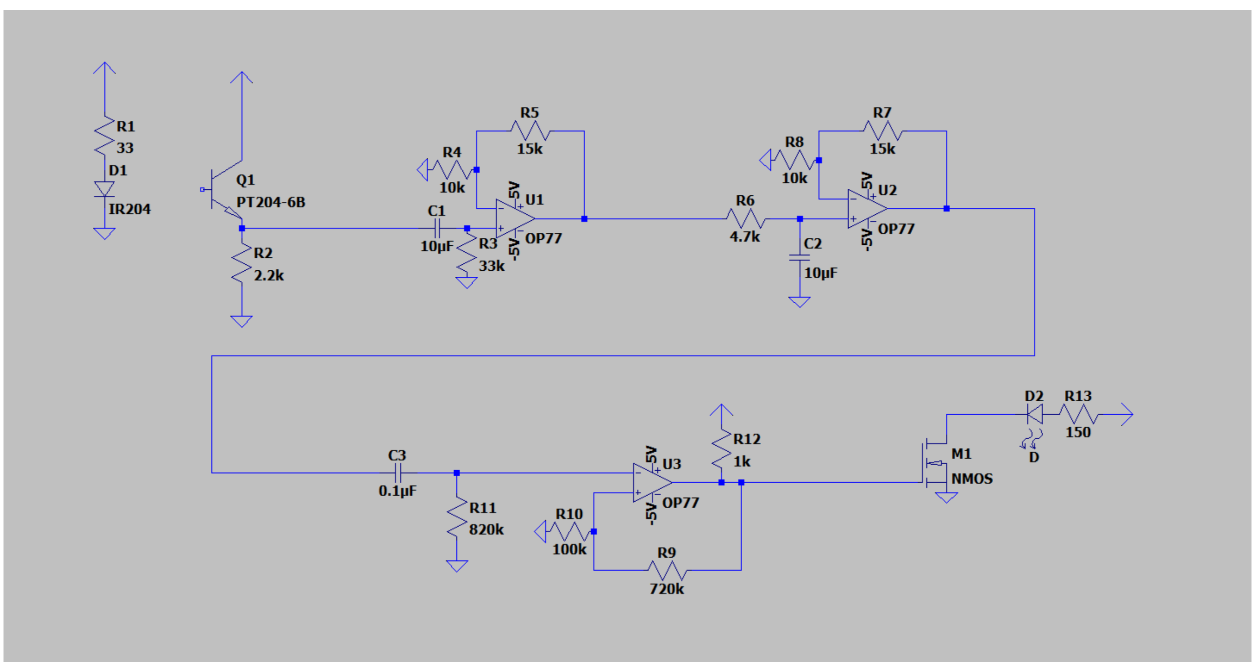


Figure 1: Circuit diagram of final circuit layout

Plots, Images, and Screenshots:

Figure 2: Oscilloscope screenshot of raw sensor output

Above you can see the raw output of our optical sensor system, where the small spikes correspond to each heartbeat.

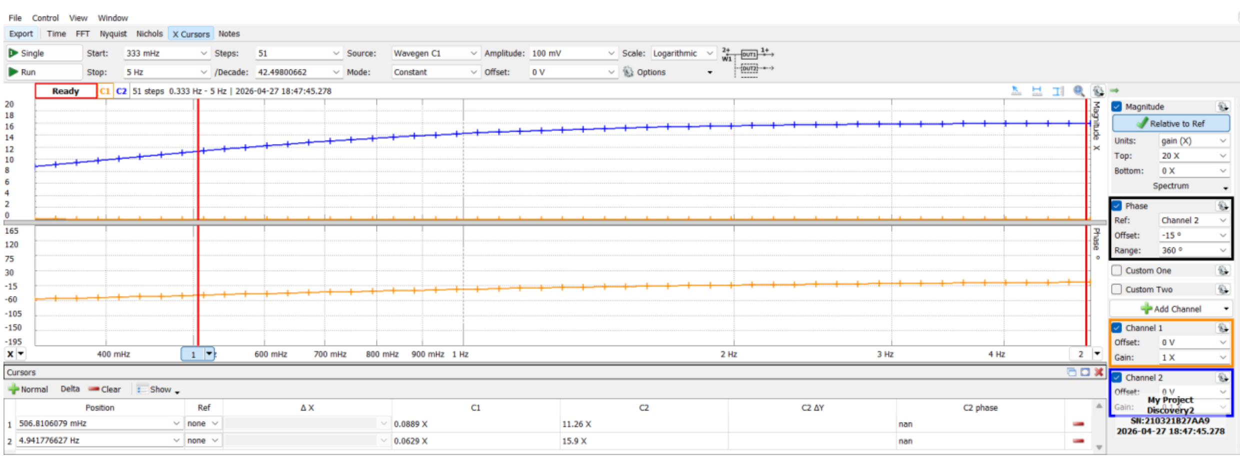


Figure 3: FRA of high pass filter

Above is our FRA screenshot of our high pass filter. This matches our intended behavior as the gain lies at 16 at the end and at the desired cutoff frequency, our gain lies at 1/(2)^0.5 times of our desired gain of 16.

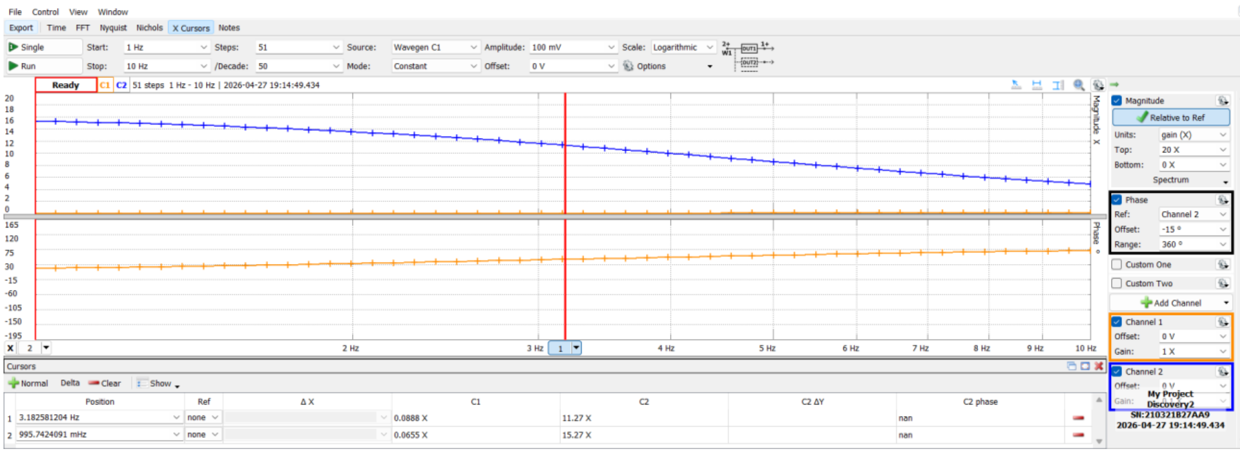


Figure 4: FRA of low pass filter

Above is our FRA of our low pass filter, where the gain also lies at our desired value of 16. Additionally, at the cutoff frequency, the gain demonstrates the same behavior where it is 1/(2)^.5 times of the total system gain.

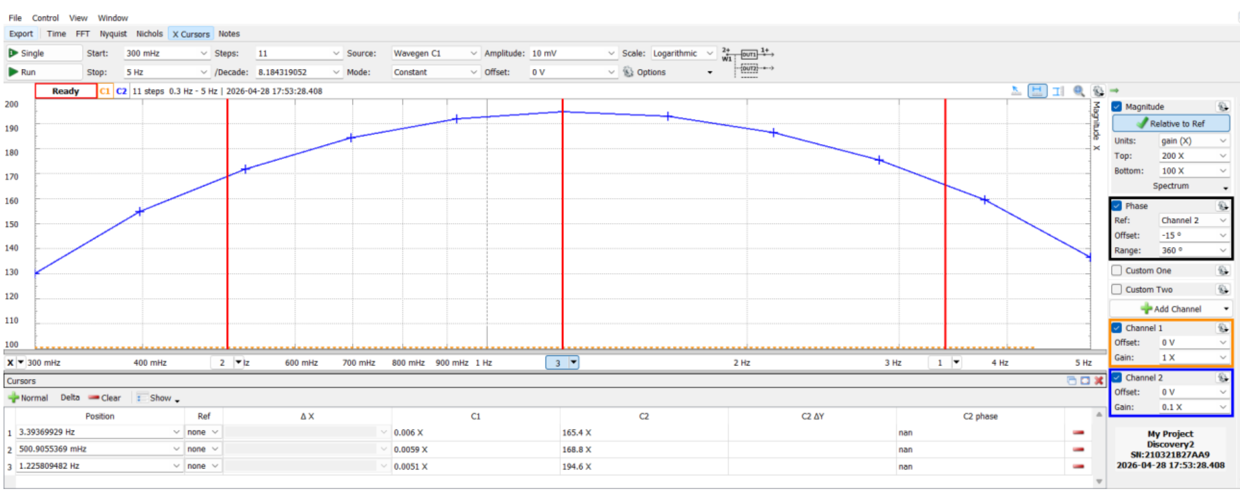


Figure 5: FRA of band pass filter

Above is our FRA of both filters, creating a band pass filter. Here we can see how the shape of the graph matches the typical shape of a band pass filter and how the gain at the center frequency is the highest.

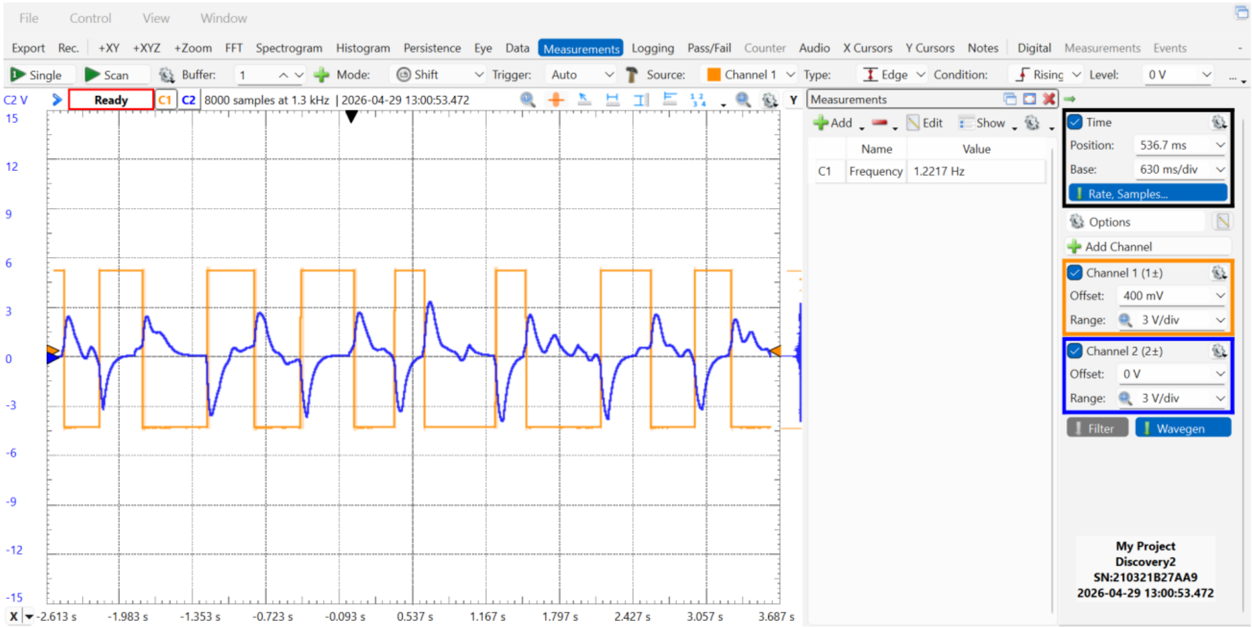


Figure 6: Analog to digital conversion oscilloscope screenshot

Above is a screenshot showing a working version of our analog to digital conversion, where the heartbeat is actually being converted into a digital binary output. Additionally, it can be shown that the frequency is at an appropriate value for a heartbeat.

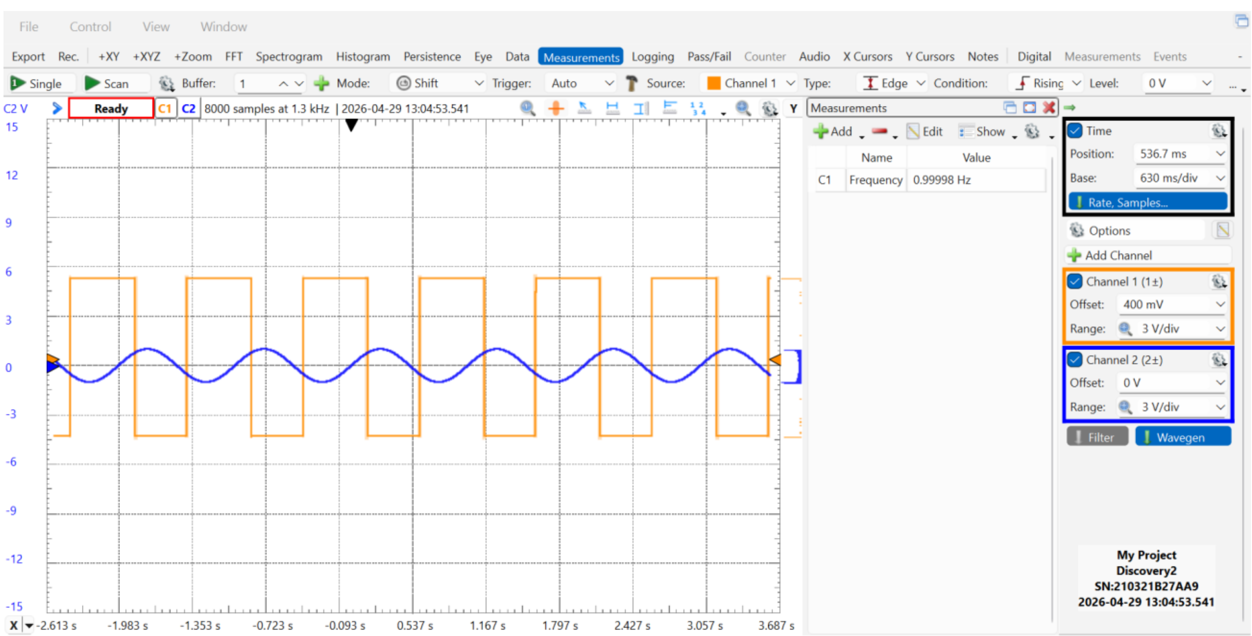


Figure 7: Analog to digital proof of concept oscilloscope screenshot

Above is another screenshot of our working analog to digital conversion, where a sine wave is being converted to a digital signal.

Conclusion

Overall, this lab experiment can be considered a success. We were able to successfully construct our circuit and see it working, both through our oscilloscope as well as visually through the LED. I believe I learned a lot about op-amps and analog design through this experiment, and it really tied together a lot of the concepts that were taught throughout this course. All of the separate subsystems of this design worked, as seen through the results section, and the final comparator output also showed that the overall system worked well. The separate subsystems were accurate on their own and were integrated successfully, allowing for behavior that matched well with the design specifications.